

Bif401p
Final Quiz File

Bif401p - Bioinformatics I (Practical) (Final Quiz)

Question # 1 of 25 (Start time: 03:13:37 PM, 11 February 2023)

Sequence Alignment helps scientists

Select the correct option

☒	All of these	correct
☐	To trace out evolutionary relationship	
☐	To predict new members of gene families	
☐	To infer the functions of newly synthesized genes	

Which of the following statement is not true about BLAST program?

Select the correct option

☐	The programs are organized based on the type of nucleotide sequences OR nucleotide sequence to be translated	
☐	BLAST is not based on heuristic searching methods	Correct
☐	The programs are organized based on the type of query sequences	
☐	The BLAST web server has been designed in such a way to simplify the task of program selection	

Reload Math Equation

CSM stands for

Hint: Protein Data Bank (PDB)

Select the correct option

- ☐ Computer Structure Management
- ☐ Computer System Management
- ☐ Computer System Models
- ☒ Computer Structure Models

BLASTX searches a protein sequence database using protein sequences as queries?

Select the correct option

- ☐ TRUE
- ☒ FALSE

What is the main function of UniProt Metagenomic and Environmental Sequences (UniMES)?

Select the correct option

- ☐ To maintain a vast collection of bioinformatics databases and services
- ☒ To store sequences which are recovered directly from environmental samples
- ☐ To provide a reference set of clusters for all UniProtKB sequences
- ☐ To provide comprehensive protein sequence and functional information

Reload Math Eq

What is the main function of UniProt Archive (UniParc)? Total Marks: 1

Select the correct option

☐ To provide compressive protein sequence and functional information

☐ To maintain a vast collection of bioinformatics database and services

☐ To provide a reference set of clusters for all UniProtKB sequences

☒ To capture all publicly available protein sequence data and contains all the protein sequences from the main publicly available protein sequence databases

[Reload Math Equations](#)

[Click to Save Answer & Move to Next Question](#)

Question # 7 of 25 (Start time: 03:18:54 PM, 11 February 2023)

A phylogenetic tree shows evolutionary relationships among _____ that have a common ancestor.

Select the correct option

☐ Genomes

☐ Species

☒ All these

☐ Genes

BIF401P - Bioinformatics I (Practical) (Final Lab Quiz) Quiz Start Time: 03:13 PM, 11 February 2023

Question # 8 of 25 (Start time: 03:19:43 PM, 11 February 2023) Total Marks: 1

----- a new fold recognition method aimed particularly at detecting superfamily relationships in case of psipred

Select the correct option

☐ BLAST

☒ GenTHREADER

☐ Uniprot

☐ MEMSAT 2

[Reload Math Equations](#)

Question # 9 of 25 (start time: 03:21:02 PM, 11 February 2023)

A protein having instability index 40 is considered as stable.

Select the correct option

- | | |
|----------------------------------|---------------|
| ☐ | None of these |
| ☐ | Equal to |
| ☒ | Smaller than |
| ☐ | Greater than |

----- is a recently proposed scale for measuring the structural similarity between two structures

Select the correct option

- | | |
|----------------------------------|----------|
| ☐ | C-score |
| ☐ | L-score |
| ☐ | Z-score |
| ☒ | TM-score |

Click to Save Answer

ClustalW is a recent version of Clustal with W standing for

Select the correct option

☐ Winding

☐ Wiping

☐ Weakening

☒ Weighting

..... compares protein sequence against protein databases. (Choose most relevant option)

Select the correct option

☐ blastn

☐ blastx

☐ tblastx

☒ blastp

In Target-template alignment, if the user knows the structure of the template to use for modelling, the coordinates can be uploaded in PDB format (*) together with the target protein sequence.

Total Marks:

Select the correct option

☐ False

☒ True

[Reload Math Equations](#)

PDB allows to search following 3D structures

Select the correct option

- ☐ Computed Structure Models
- ☐ CSM
- ☐ All of these
- ☐ Experimentally determined

Correct

What is the main function of UniProt Knowledgebase (UniProtKB)?

Select the correct option

- ☐ To store metagenomic and environmental sequences
- ☐ To maintain a vast collection of bioinformatics databases and services
- ☐ To provide a reference set of clusters for all UniProtKB sequences
- ☒ To provide comprehensive protein sequences and functional information

Command line versions of the ClustalW programs are available for which platform?

Select the correct option

- | | |
|-----------------------|-----------------|
| ☐ | UNIX |
| ☐ | Macintosh |
| ☐ | Window base PCs |
| ☐ | All of these |
- correct

TM-score _____ means a randomly selected protein pair with the gapless alignment taken from PDB

Select the correct option

- | | |
|----------------------------------|-------|
| ☐ | <0.18 |
| ☐ | <0.15 |
| ☐ | <0.14 |
| ☒ | <0.17 |
- Click to Save Answer & Move to Next Question

Which one of the following statements is not a step of ClustalW algorithm?

Select the correct option

Reload

- ☐ Progressively align the sequence together into each branch point of the guide tree
- ☐ Convert the sequence similarity
- ☒ Construct a tree for the order in which pairs of sequences are to be aligned and combined with previous alignments
- ☐ Compute the multiple sequence alignments for all sequences

Correct

Amino acid composition denotes of individual residues of a protein.

Select the correct option

- ☐ Total number
- ☐ Concentration
- ☐ Charge
- ☐ Strength

Correct

How does the sequence alignment help in drug discovery?

Select the correct option

- ☐ By identifying area that might be affected by mutations or drug binding
- ☐ By identifying functional domains in proteins
- ☒ All of these
- ☐ By identifying evolutionary relationship between organisms

Correct

X

How does the sequence alignment help in drug discovery?

Select the correct option

- ☐ By identifying area that might be affected by mutations or drug binding
- ☐ By identifying functional domains in proteins
- ☒ All of these
- ☐ By identifying evolutionary relationship between organisms

Correct

The human genome consists of ~ base pairs.

Select the correct option

- ☐ 2.3 billion
- ☒ 3.2 billion
- ☐ 2.3 million
- ☐ 3.2 million

Click to Save Answer & ...

In MEGA 11, you can change font size of ----- phylogenetic tree.

Select the correct option

- ☐ All these
- ☐ Branch lengths
- ☐ Bootstrap values
- ☐ Tax

Correct

Question # 23 of 25 (Start time: 03:32:31 PM, 11 February 2023)

In vivo half-life means ---- of a protein in the cell.

Select the correct option

- ☐ Replacement
- ☐ Degradation
- ☐ Replacement or degradation
- ☐ Strength

Correct

What type of sequences can be found in UniMES?

Select the correct option

- ☐ Reference set of clusters for all UniProtKB sequences
- ☐ Protein sequence and functional information
- ☐ None of these
- ☒ Metagenomic and environmental sequences

Computer structure modes are those models which are generate by

Select the correct option

☐	Experiments
☐	Manually by humans
☐	Computer algorithms
☐	Databases

Correct

Click to Save Answer & Move to Next Question

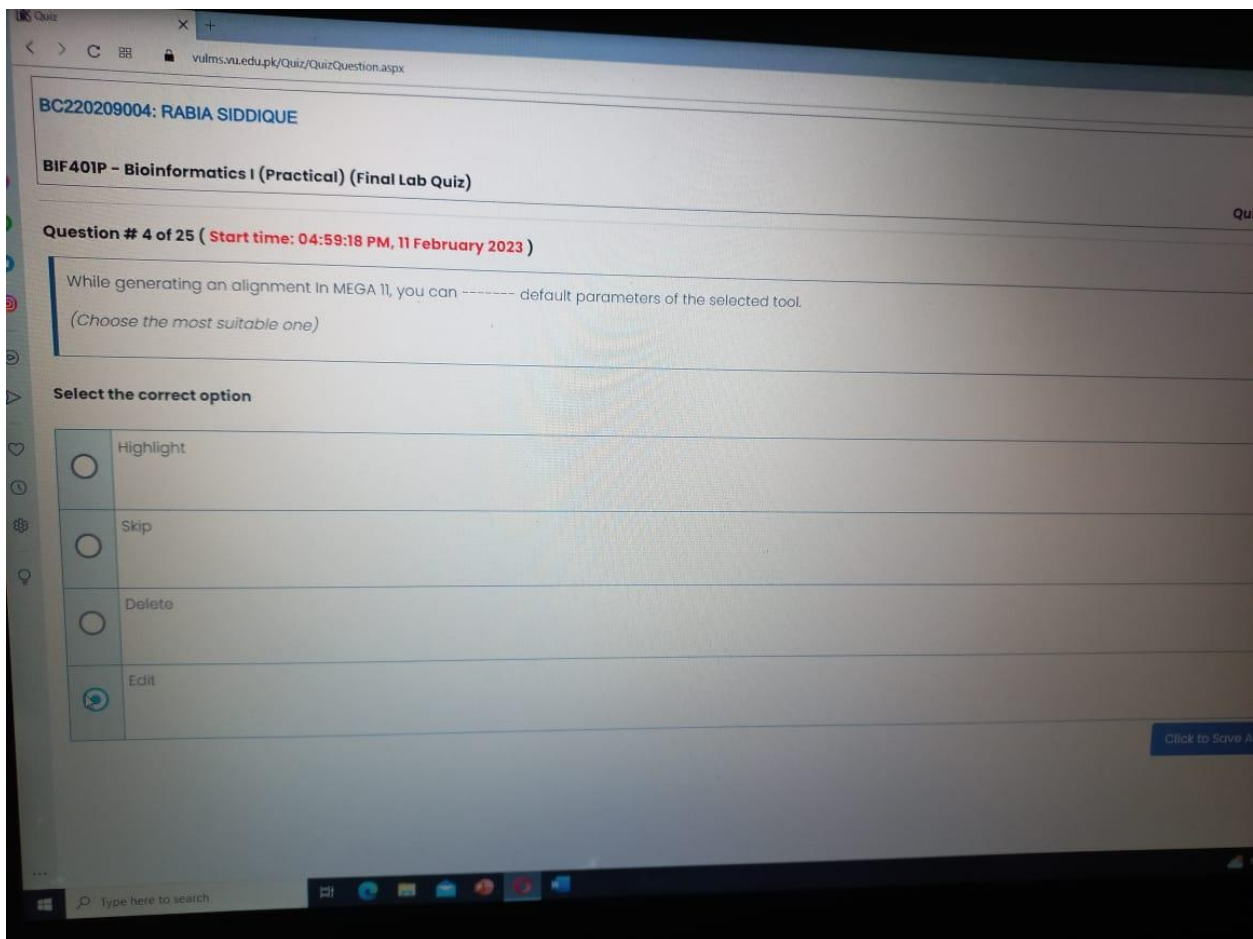
PDB allows to search following 3D structures

Select the correct option

☐	Computed Structure Models
☐	CSM
☐	All of these
☐	Experimentally determined

Correct

Click to Save Answer & Move to Next Question



What type of sequences can be found in UniMES?

Select the correct option

☐

None of these

☐

Protein sequence and functional information

☐

Reference set of clusters for all UniProtKB sequences

☒

Metagenomic and environmental sequences

Correct

_____ is incorrect statement regarding the terminologies of phylogenetics.

Select the correct option

☐

Node is ancestor of extant taxa

☐

Branches shows the lines

☐

Point joining two adjacent branches represent a node

☒

Tips of the branches shows species or genes

Correct

How does the sequence alignment help in drug discovery?

Select the correct option

- ☐ By identifying area that might be affected by mutations or drug binding
- ☐ All of these
- ☐ By identifying functional domains in proteins
- ☐ By identifying evolutionary relationship between organisms

Correct

PDB allows to search following 3D structures

Select the correct option

- ☐ All of these
- ☒ Computed Structure Models
- ☐ Experimentally determined
- ☐ CSM

Correct

Question # 6 of 25 (Start time: 09:08:40 PM, 12 February 2023)

----- is a short sequence that contains one or more conserved residues, which allow the protein to bind to a ligand and carry out a catalytic activity.
Hint: InterPro

Select the correct option

- ☒ Conserved site
- ☐ Active site
- ☐ PTM site
- ☐ Binding site

Correct

Question # 7 of 25 (Start time: 09:07:48 PM, 12 February 2023)

----- compares protein sequence against protein databases. (Choose most relevant option)

Select the correct option

- ☐ blastx
- ☒ blastp
- ☐ blastn
- ☐ tblastx

In an alignment all sequences have length.

Select the correct option

- ☒ Equal
- ☐ Multiple of 2
- ☐ None of these
- ☐ Different

Correct

Computer structure modes are those models which are generate by

Select the correct option

- ☐ Manually by humans
- ☐ Experiments
- ☒ Databases
- ☐ Computer algorithms

Correct

What is the main function of UniProt Knowledgebase (UniProtKB)?

Select the correct option

- ☐ To store metagenomic and environmental sequences
- ☐ To provide a reference set of clusters for all UniProtKB sequences
- ☐ To provide comprehensive protein sequences and functional information
- ☒ To maintain a vast collection of bioinformatics databases and services

Correct

----- a new fold recognition method aimed particularly at detecting superfamily relationships in case of psipred

Select the correct option

- ☐ GenTHREADER
- ☐ BLAST
- ☒ Uniprot
- ☐ MEMSAT 2

Correct

_____ is a recently proposed scale for measuring the structural similarity between two structures

Select the correct option

☒ Z-score

☐ L-score

☐ TM-score

☐ C-score

Correct

Sequence Alignment helps scientists

Select the correct option

☐ To infer the functions of newly synthesized genes

☐ To trace out evolutionary relationship

☒ All of these

☐ To predict new members of gene families

In vivo half-life means ---- of a protein in the cell.

Select the correct option

- ☐ Replacement
- ☐ Strength
- ☐ Replacement or degradation
- ☒ Degradation

Correct

The human genome consists of - base pairs.

Select the correct option

- ☐ 2.3 million
- ☐ 3.2 million
- ☒ 3.2 billion
- ☐ 2.3 billion

CSM stands for

Hint: Protein Data Bank (PDB)

Select the correct option

- | | |
|----------------------------------|-------------------------------|
| ☐ | Computer Structure Management |
| ☒ | Computer Structure Models |
| ☐ | Computer System Management |
| ☐ | Computer System Models |

What is the main function of UniProt Archive (UniParc)?

Select the correct option

- | | |
|----------------------------------|--|
| ☒ | To provide a reference set of clusters for all UniProtKB sequences |
| ☐ | To provide compressive protein sequence and functional information |
| ☐ | To maintain a vast collection of bioinformatics database and services |
| ☐ | To capture all publicly available protein sequence data and contains all the protein sequences |

Correct

In MEGA 11, you can change font size of ----- phylogenetic tree.

Select the correct option

☒	All these
☐	Tax
☐	Bootstrap values
☐	Branch lengths

GMQE (Global Model Quality Estimation) is expressed as a number between -----

Select the correct option

☒	0 and 1
☐	Greater than 1
☐	0.5 to 1
☐	Less than 0

GenBank, the nucleic acid sequence database is maintained by

Select the correct option

- ☐ European Molecular Biology Laboratory (EMBL)
- ☒ National Centre for Biotechnology Information (NCBI)
- ☐ DNA Database of Japan (DDBJ)
- ☐ UniProt/ Swiss-Prot Database

Which of the following statements about the global alignment algorithm is incorrect?
(Recall the concepts of Needleman-Wunsch Algorithm)

Select the correct option

- ☐ First row and First column are set to Zero "0"
- ☐ In initialization step, the first row and first column are subject to gap penalty
- ☒ Score can be negative
- ☐ In trace-back step, beginning is with the cell at the lower right of the matrix, and it ends at top

Correct